Vendor Performance and Inventory Optimization

Data Analysis Project Report

SQL | Python (Pandas, Seaborn, SciPy) | Power BI

Mohammed Ayan

B.Tech, Artificial Intelligence and Machine Learning

mohammedayan.aiml@gmail.com | GitHub | LinkedIn

Contents

Contents	2
1. Executive Summary	3
2. Business Problem and Objectives	4
3. Data and Architecture	5
4. Data Preparation and Quality	6
5. Exploratory Data Analysis	8
6. Analysis and Findings	10
7. Power BI Dashboard	16
8. Recommendations	17
9. Limitations and Next Steps	18
10. Tools and Technologies	18
11. Conclusion	18

1. Executive Summary

This project analyses vendor, purchasing, sales and inventory data of a large multi-store beverage retailer to answer a practical question: **which vendors and brands create value, and where is money being tied up or lost?** Raw transaction tables (more than 12.8 million sales rows and 2.3 million purchase rows) were consolidated with SQL into one vendor-brand summary table, cleaned and explored in Python, and presented in an interactive Power BI dashboard.

Total Sales	Total Purchases	Gross Profit	Unsold Capital	Top-10 Vendor Share of Purchases
$441.41M	**$307.34M**	**$134.07M**	**$2.71M**	**65.69%**

Key findings

- **Purchasing is concentrated.** The ten largest vendors account for 65.69% of all purchase spend; Diageo North America alone is 16.30%.
- **Capital is locked in slow stock.** About $2.71M (net) sits in unsold inventory. The largest holders are Diageo ($0.72M), Jim Beam ($0.55M) and Pernod Ricard ($0.47M).
- **Some vendors barely sell what they buy.** Ten vendors have a stock turnover below 1; the weakest is Alisa Carr Beverages at 0.62.
- **Low-selling items earn higher margins.** Average margin is 41.57% for the lowest-selling quartile versus 31.18% for the top quartile (Welch t-test, $p < 0.0001$).
- **Bulk buying lowers unit cost.** Large orders average $10.78 per unit versus $39.07 for small orders, about 72% lower (partly a product-mix effect, see Section 6).
- **Pricing and promotion candidates exist.** A group of brands sit below the 15th percentile of sales ($560) yet above the 85th percentile of margin (64.97%).

Recommendations are given in Section 8: reduce vendor dependence, right-size purchasing for slow movers, promote or re-price low-sales/high-margin brands, and use volume purchasing only where demand supports it.

2. Business Problem and Objectives

A retailer that buys from hundreds of vendors needs to know which relationships are profitable, which are risky, and where inventory ties up cash. Purchasing, invoice, price and sales data live in separate tables, so no single view of vendor performance existed. The project builds that view and uses it to answer six business questions.

#	Business question	Method
1	Which brands need promotional or pricing adjustments (low sales, high margin)?	Percentile thresholds, scatter plot
2	Which vendors and brands deliver the highest sales?	Group-by aggregation, ranking
3	How dependent is procurement on the top vendors?	Purchase contribution, Pareto chart
4	Does bulk purchasing reduce unit price?	Order-size tiers, box plot
5	Which vendors have low inventory turnover, and how much capital is locked in unsold stock?	Stock turnover, unsold inventory value
6	Do top and low performers differ significantly in profit margin?	95% confidence intervals, Welch t-test

3. Data and Architecture

Source data was loaded into a SQLite database (**inventory.db**) containing six raw tables plus the consolidated summary table created in this project.

Table	Records	Contents
sales	12,825,363	Store-level sales: quantity, dollars, price, date, excise tax, vendor
purchases	2,372,474	Purchase lines: quantity, dollars, purchase price, vendor
purchase_prices	12,261	Brand list price, purchase price, volume, classification
vendor_invoice	5,543	Invoices: PO and pay dates, quantity, dollars, freight, approval
begin_inventory	206,529	Opening on-hand stock by store and brand
end_inventory	224,489	Closing on-hand stock by store and brand
vendor_sales_summary	10,692	**Created in this project**: one row per vendor and brand

Pipeline

- **Ingest** raw files into SQLite using a small ingestion module with logging.
- **Explore** each table to identify keys (vendor number, brand) and the variables worth keeping.
- **Consolidate** with SQL: Common Table Expressions (CTEs) aggregate freight per vendor, purchases per vendor-brand and sales per vendor-brand, then join them into one summary table. Pre-aggregating avoids re-running expensive joins on millions of rows every time the dashboard refreshes.
- **Engineer metrics** in pandas and write the finished table back to the database and to CSV for Power BI.
- **Analyse and visualise** in Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy), then build the Power BI dashboard.

Engineered metrics

Metric	Definition
Gross Profit	Total Sales Dollars - Total Purchase Dollars
Profit Margin (%)	Gross Profit / Total Purchase Dollars x 100 (margin on purchase cost, as defined in the notebook)
Stock Turnover	Total Sales Quantity / Total Purchase Quantity
Sales-to-Purchase Ratio	Total Sales Dollars / Total Purchase Dollars
Purchase Contribution (%)	Vendor purchase dollars / total purchase dollars x 100
Unsold Inventory Value	(Purchase Quantity - Sales Quantity) x Purchase Price

4. Data Preparation and Quality

Summary statistics on the 10,692-row vendor-brand table exposed several issues that had to be handled before analysis:

- **Missing sales:** 178 vendor-brand rows had no sales data (products bought but never sold); these were filled with 0.
- **Losses:** gross profit reached as low as -$52,002.78, meaning some items sold below cost.
- **Undefined margin:** rows with zero sales produced an infinite negative profit margin through division by zero.
- **Whitespace:** vendor names carried trailing spaces and were trimmed so the same vendor is not counted twice.
- **Data types:** the Volume column was stored as text and converted to numeric.

For the performance analysis, the table was filtered to records with positive gross profit, positive profit margin and positive sales quantity, leaving **8,564 records**. This keeps ratios meaningful, but it also means averages describe profitable, selling items only (see Limitations).

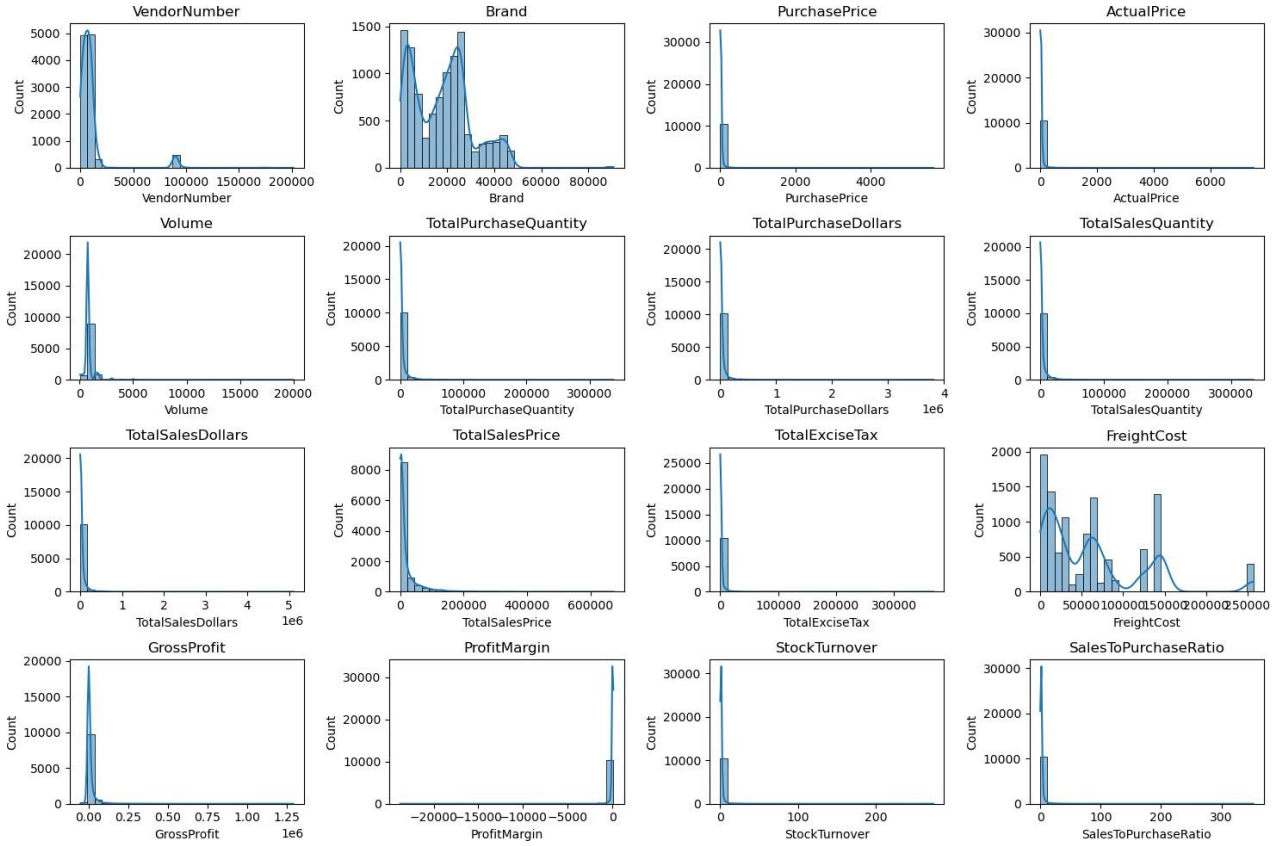


Figure 1. Distributions of numeric columns before filtering: strongly right-skewed with long tails.

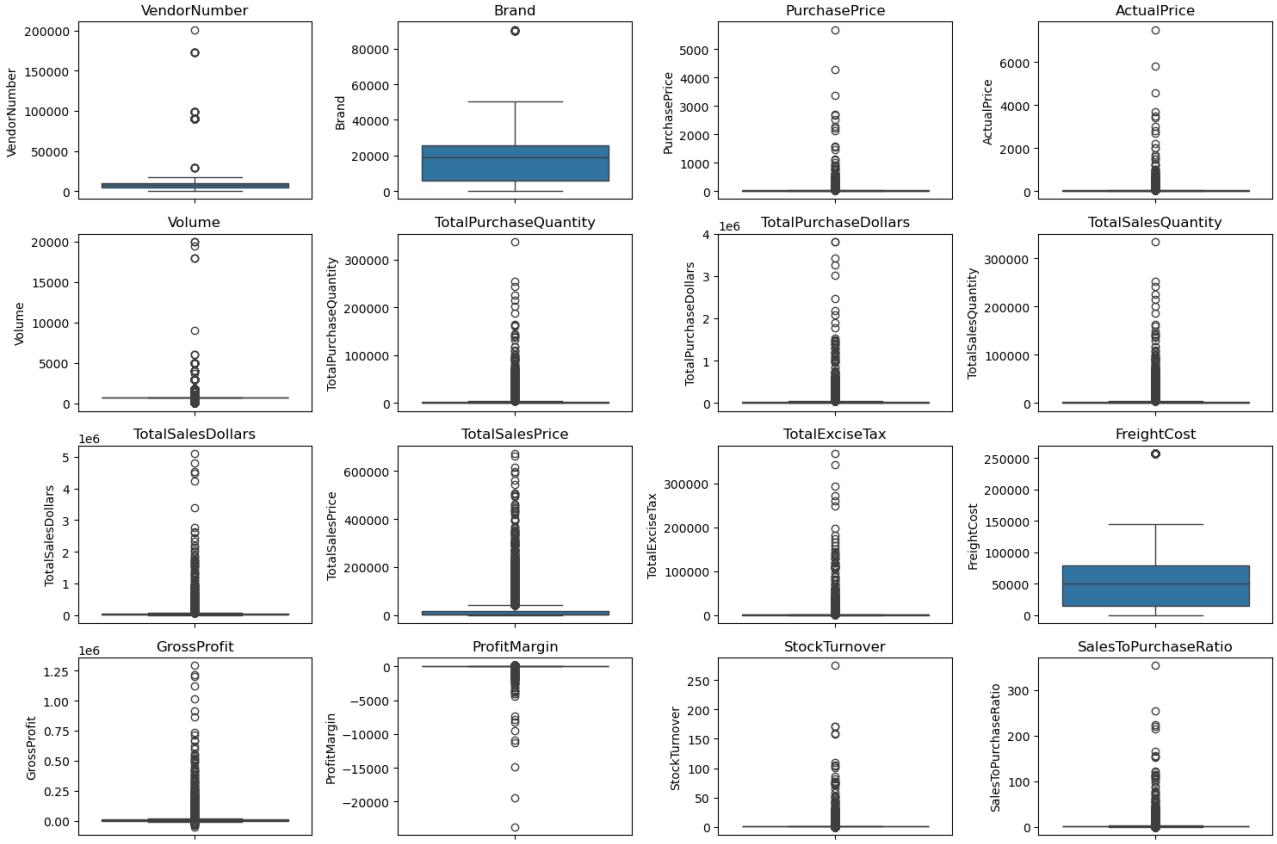


Figure 2. Box plots show heavy outliers, e.g. purchase price up to $5,681.81 and freight cost up to $257,032.07.

5. Exploratory Data Analysis

After filtering, distributions are cleaner but remain skewed, which is typical of sales data where a few products and vendors dominate. The count plots show which vendors and products appear most often in the cleaned data.

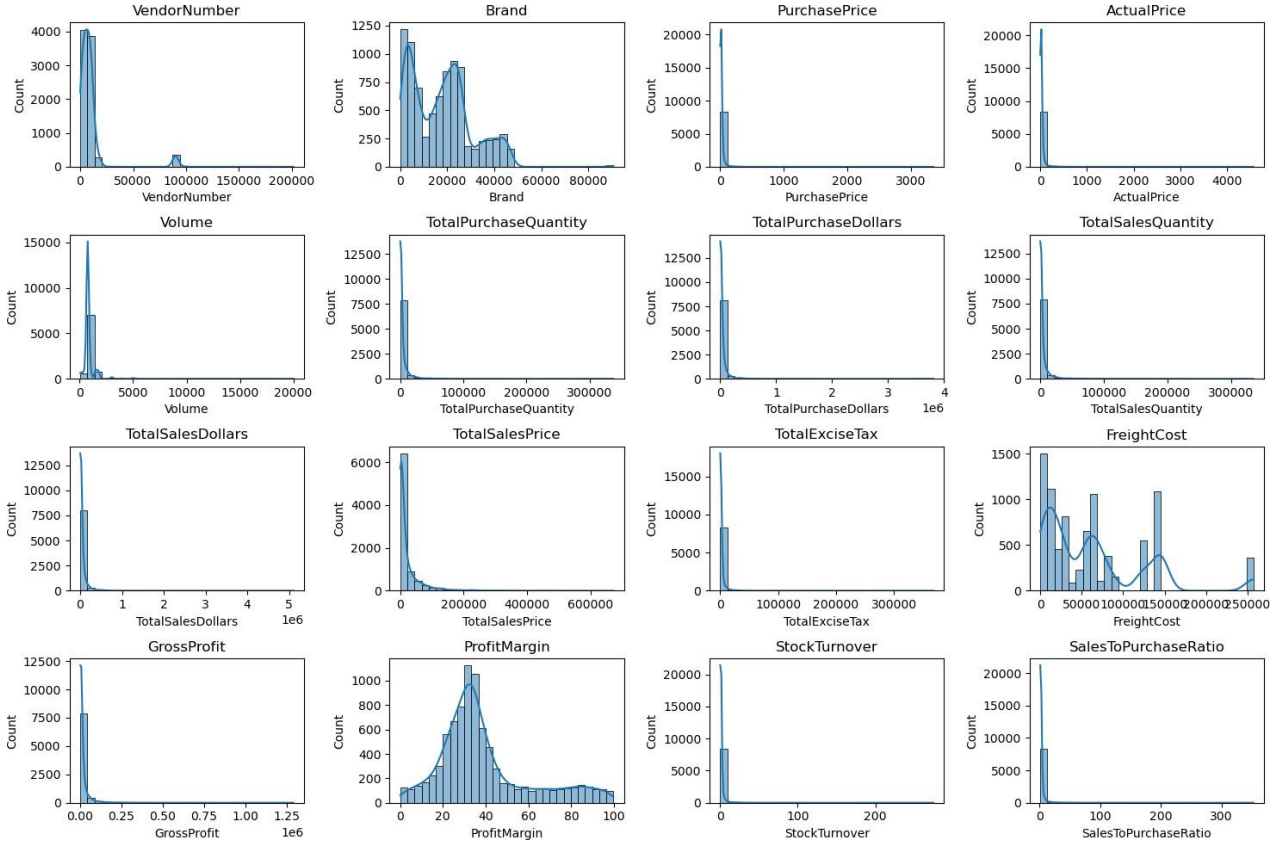


Figure 3. Distributions of numeric columns after filtering.

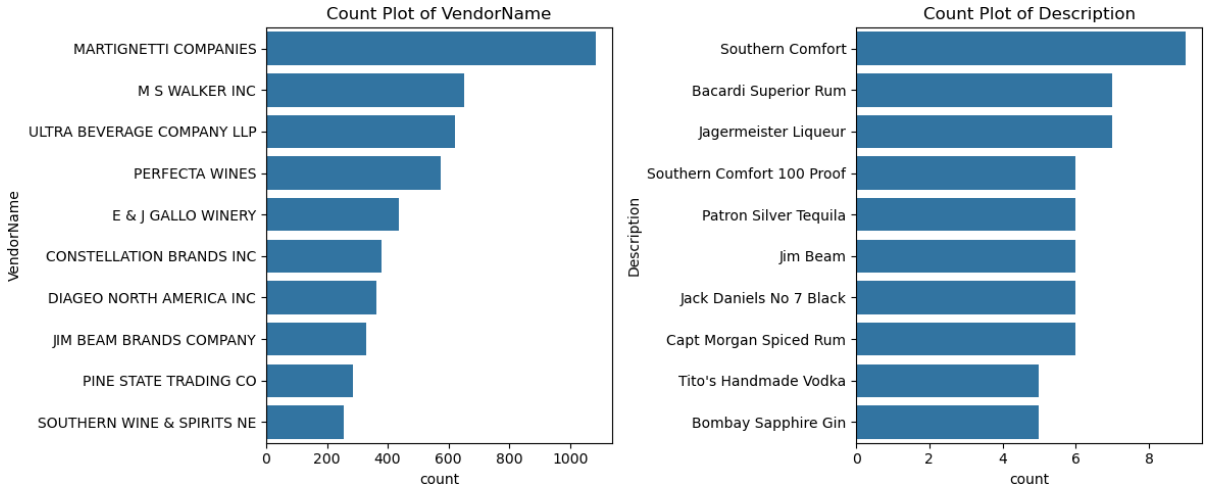


Figure 4. Most frequent vendors and product descriptions in the cleaned data.

Correlation analysis

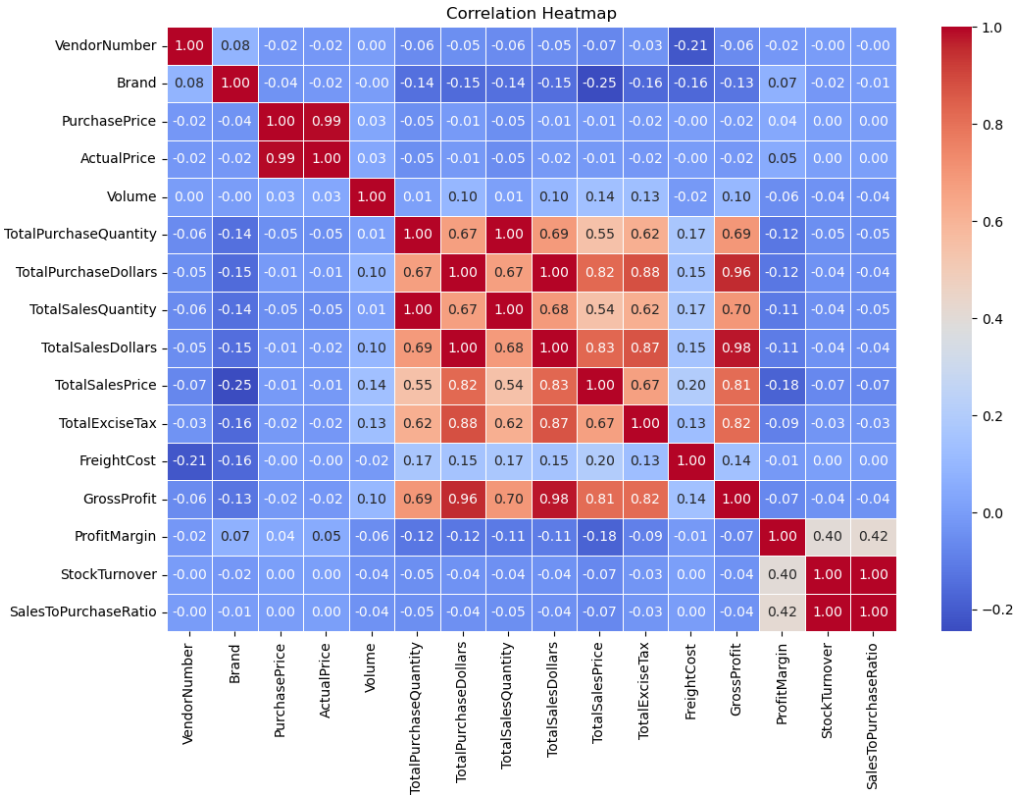


Figure 5. Correlation heatmap of numeric variables.

- Purchase quantity and sales quantity are almost perfectly correlated (about 0.999): what is bought is, in volume terms, largely what is sold.
- Purchase price has almost no linear relationship with sales dollars (-0.012) or gross profit (-0.016).
- Profit margin is weakly negatively correlated with sales price (-0.179): higher-priced items do not earn higher margins here.
- Stock turnover is only weakly related to gross profit (-0.038) and margin (-0.055): selling faster does not by itself mean earning more.

6. Analysis and Findings

6.1 Brands needing promotional or pricing adjustment

Brands were aggregated by description. The **15th percentile of total sales ($560.30)** and the **85th percentile of average profit margin (64.97%)** define a target group: brands that sell little but earn a lot per sale. These are candidates for promotion, better placement, or price testing, because a small volume increase could add meaningful profit.

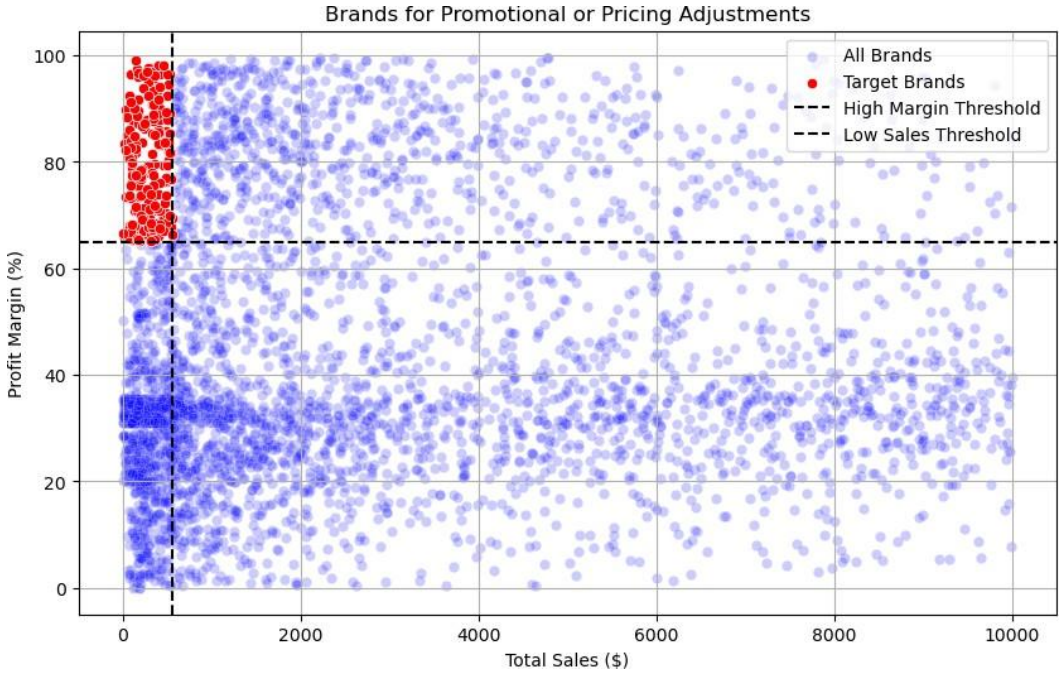


Figure 6. Brands with low sales and high margin (red) relative to all brands, thresholds shown as dashed lines.

Examples from the target group include Crown Royal Apple (sales $27.86, margin 89.8%), Concannon Glen Ellen Wh Zin ($15.95, 83.4%) and Santa Rita Organic Svgn Bl ($9.99, 66.5%).

6.2 Top vendors and brands by sales

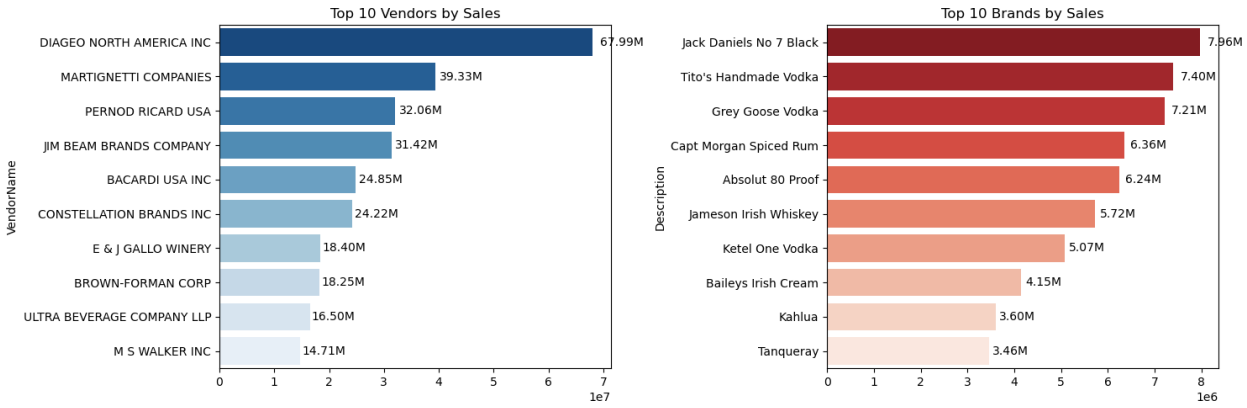


Figure 7. Top 10 vendors and top 10 brands by sales dollars.

Rank	Vendor	Sales	Brand	Sales
1	Diageo North America	$67.99M	Jack Daniels No 7 Black	$7.96M
2	Martignetti Companies	$39.33M	Tito's Handmade Vodka	$7.40M
3	Pernod Ricard USA	$32.06M	Grey Goose Vodka	$7.21M
4	Jim Beam Brands Company	$31.42M	Capt Morgan Spiced Rum	$6.36M
5	Bacardi USA Inc	$24.85M	Absolut 80 Proof	$6.24M
6	Constellation Brands Inc	$24.22M	Jameson Irish Whiskey	$5.72M
7	E & J Gallo Winery	$18.40M	Ketel One Vodka	$5.07M
8	Brown-Forman Corp	$18.25M	Baileys Irish Cream	$4.15M
9	Ultra Beverage Company LLP	$16.50M	Kahlua	$3.60M
10	M S Walker Inc	$14.71M	Tanqueray	$3.46M

Diageo leads by a wide margin, selling about 1.7 times as much as the second-ranked vendor. Vodka and whiskey brands dominate the brand ranking.

6.3 Procurement dependence on top vendors

Purchase contribution measures each vendor's share of total purchase spend. The top ten vendors together account for **65.69%** of all purchases, and Diageo North America alone for **16.30%** ($50.10M).

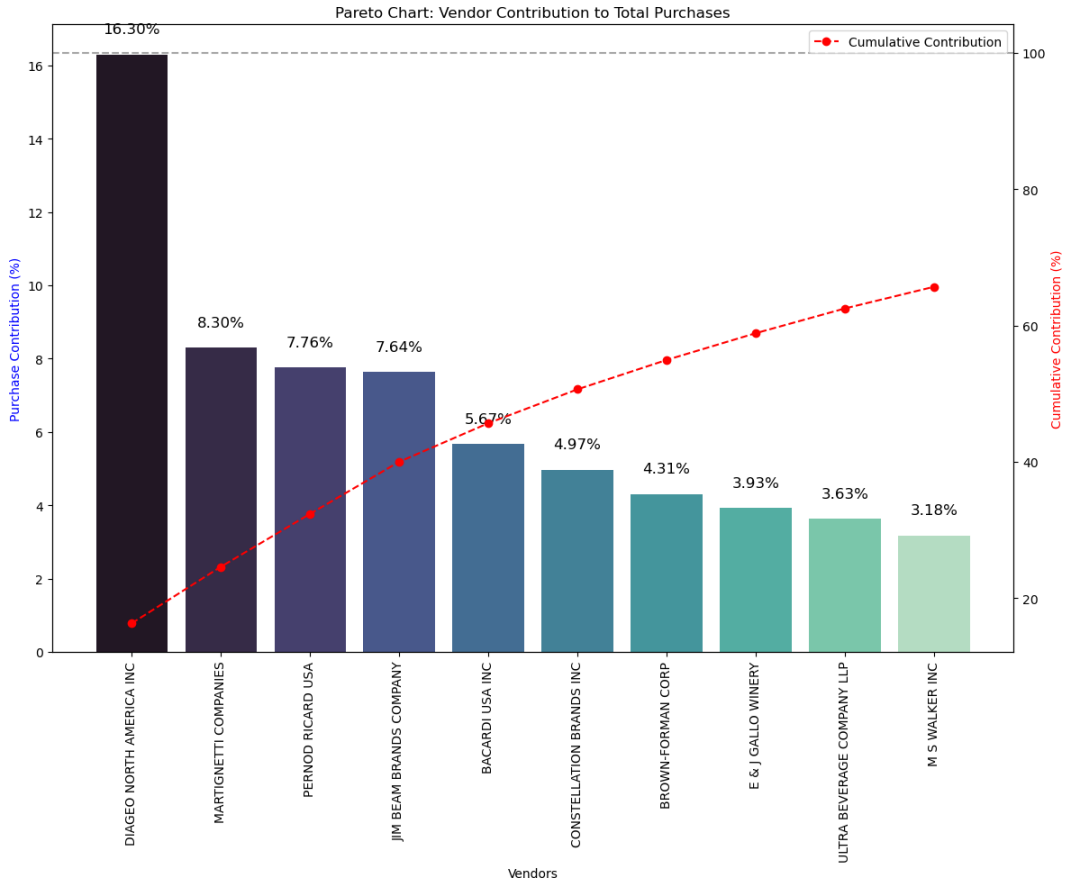


Figure 8. Pareto chart: purchase contribution of the top 10 vendors with cumulative share (red line).

Vendor	Purchases	Gross profit	Share	Cumulative
Diageo North America	$50.10M	$17.89M	16.30%	16.30%
Martignetti Companies	$25.50M	$13.83M	8.30%	24.60%
Pernod Ricard USA	$23.85M	$8.21M	7.76%	32.36%
Jim Beam Brands Company	$23.49M	$7.93M	7.64%	40.00%
Bacardi USA Inc	$17.43M	$7.42M	5.67%	45.67%
Constellation Brands Inc	$15.27M	$8.95M	4.97%	50.64%
Brown-Forman Corp	$13.24M	$5.01M	4.31%	54.95%
E & J Gallo Winery	$12.07M	$6.33M	3.93%	58.88%
Ultra Beverage Company LLP	$11.17M	$5.34M	3.63%	62.51%
M S Walker Inc	$9.76M	$4.94M	3.18%	65.69%

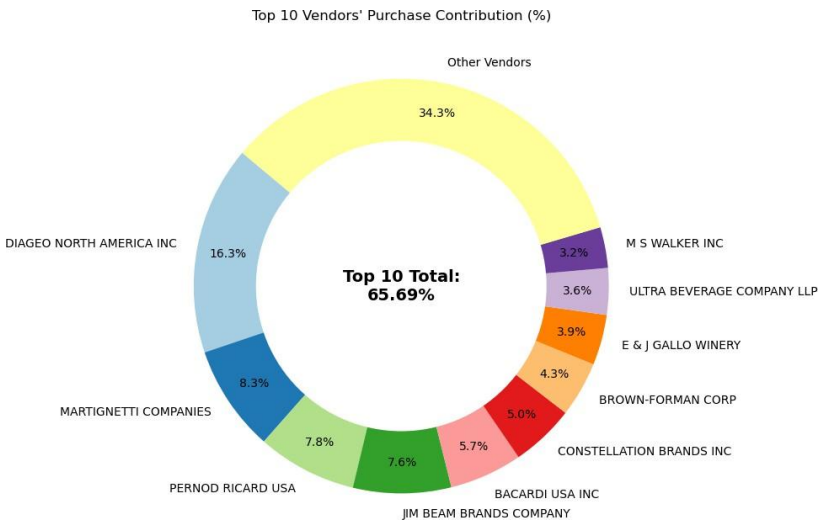


Figure 9. Purchase share of the top 10 vendors and all other vendors.

Note: the donut in the Power BI dashboard shows each vendor's share within the top 10 (for example, Diageo 24.81% = 16.30 / 65.69), whereas the figures above are shares of total purchases.

6.4 Does bulk purchasing reduce unit price?

Records were split into three equal-sized groups (Small, Medium, Large) by total purchase quantity, and the average unit purchase price was compared.

Order size	Average unit purchase price
Small	$39.07
Medium	$15.49
Large	$10.78

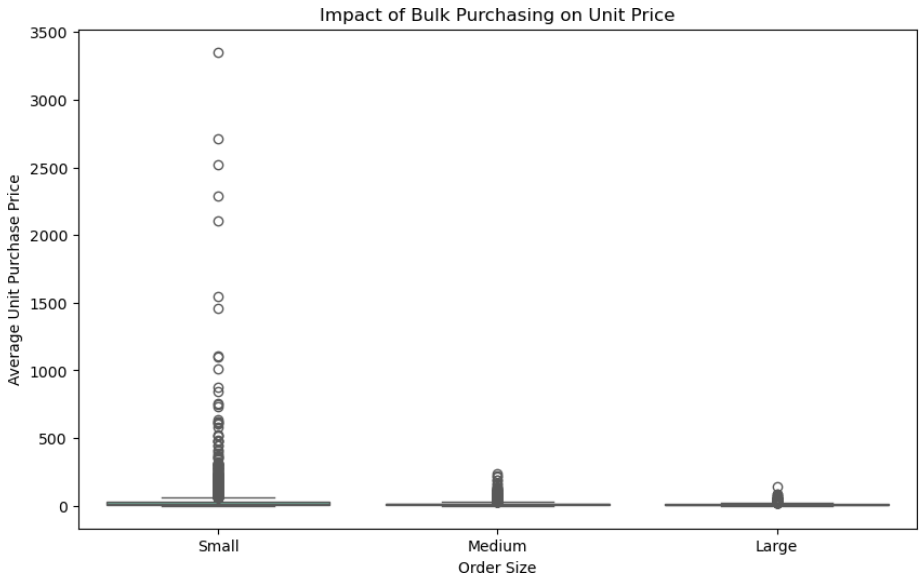


Figure 10. Unit purchase price by order size.

Large orders carry a unit price about 72% lower than small orders, which supports volume purchasing. Interpret this carefully: small-volume items are often premium products, so part of the gap reflects product mix rather than a pure volume discount.

6.5 Low inventory turnover and locked capital

Stock turnover (sales quantity divided by purchase quantity) below 1 means a vendor sold less than it bought, leaving excess stock. The ten vendors with the lowest average turnover are:

Vendor	Stock turnover	Vendor	Stock turnover
Alisa Carr Beverages	0.615	Centeur Imports LLC	0.774
Highland Wine Merchants LLC	0.708	Smoky Quartz Distillery LLC	0.784
Park Street Imports LLC	0.751	Tamworth Distilling	0.797
Circa Wines	0.756	The Imported Grape LLC	0.808
Dunn Wine Brokers	0.766	Walpole Mtn View Winery	0.821

Unsold inventory value was calculated as (purchase quantity - sales quantity) x purchase price. The total is **$2.71M**. Vendors with the most capital locked in unsold stock:

Vendor	Unsold inventory value
Diageo North America	$722,209
Jim Beam Brands Company	$554,666
Pernod Ricard USA	$470,626
William Grant & Sons Inc	$401,961
E & J Gallo Winery	$228,283
Sazerac Co Inc	$198,436
Brown-Forman Corp	$177,734
Constellation Brands Inc	$133,618
Moet Hennessy USA Inc	$126,478
Remy Cointreau USA Inc	$118,598

*The biggest vendors also hold the most unsold stock, simply because they carry the largest volumes. The $2.71M total is a **net** figure: vendors who sold more than they bought in the period (turnover above 1, drawing on opening stock) show negative values that offset part of the positive balances.*

6.6 Are margins different for top and low performers?

Records in the top quartile of sales were compared with those in the bottom quartile. 95% confidence intervals for mean profit margin do not overlap:

Group	Mean margin	95% confidence interval
Top performers (sales in top 25%)	31.18%	30.74% to 31.61%
Low performers (sales in bottom 25%)	41.57%	40.50% to 42.64%

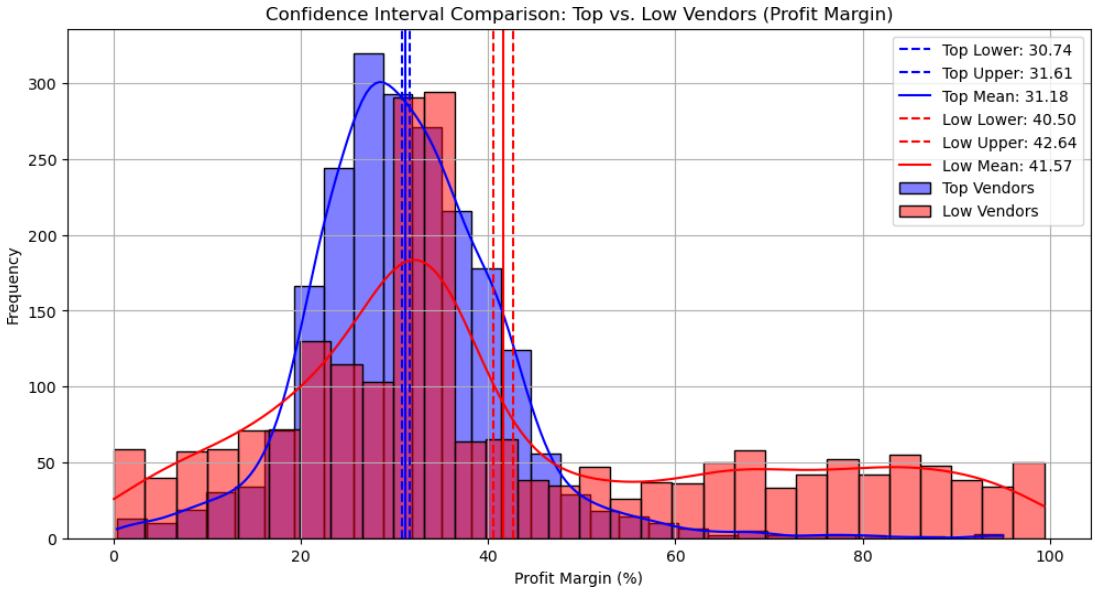


Figure 11. Profit-margin distributions of top and low performers with 95% confidence intervals.

Hypothesis test. H0: mean margins are equal. H1: they differ. A two-sample Welch t-test gave t = -17.67 and p < 0.0001, so H0 is rejected: the difference is statistically significant. Low-selling items earn higher margins, probably through premium pricing or small volumes, while high sellers compete on volume. Top sellers could test selective price rises or bundling; low sellers need better marketing, pricing or distribution to turn margin into volume.

7. Power BI Dashboard

The summary table was loaded into Power BI to give managers a self-service view of the same analysis.

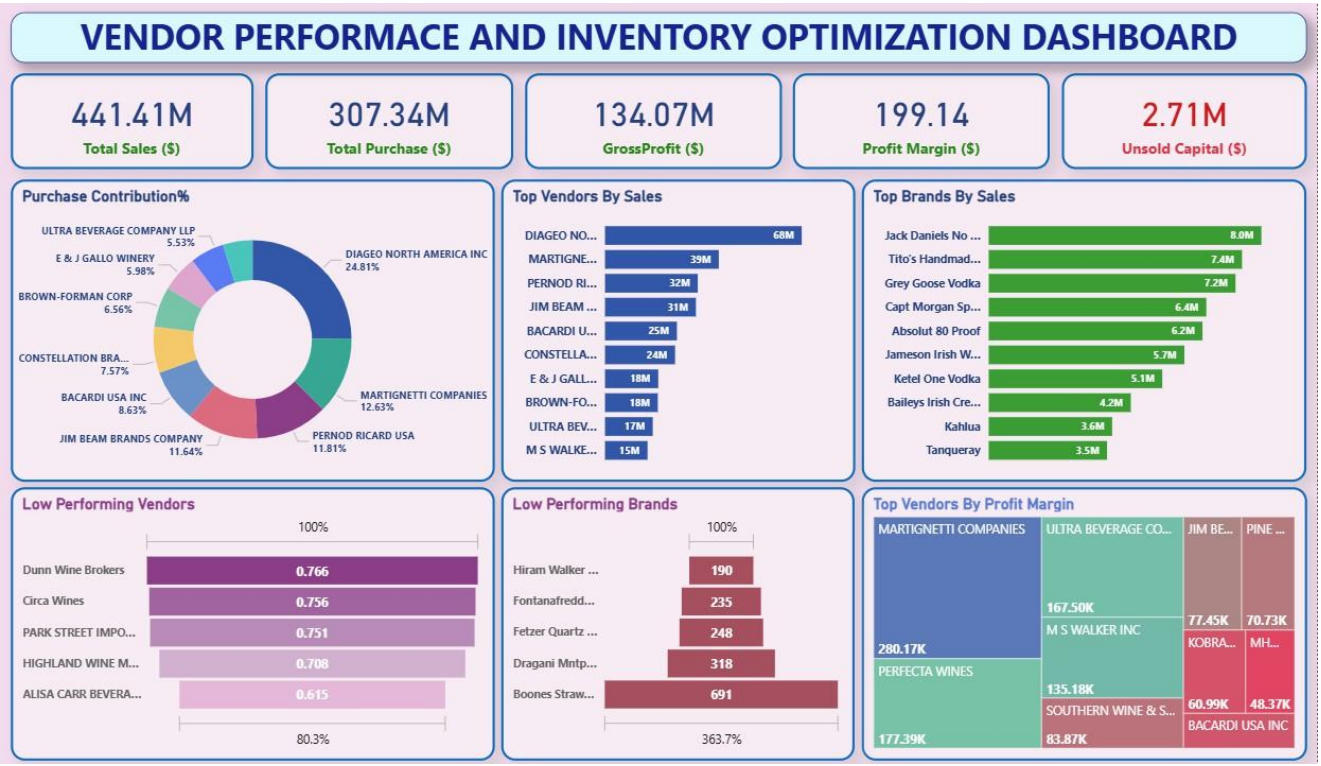


Figure 12. Vendor Performance and Inventory Optimization dashboard.

Dashboard element	What it shows
KPI cards	Total sales, total purchases, gross profit, profit margin and unsold capital
Purchase Contribution %	Each top-10 vendor's share of purchase spend (within the top 10)
Top Vendors / Top Brands by Sales	Ranked bar charts of the highest sellers
Low Performing Vendors	Vendors with the lowest stock turnover (0.615 to 0.766 shown)
Low Performing Brands	Brands flagged as low-sales, high-margin
Top Vendors by Profit Margin	Treemap of vendors by profit contribution

8. Recommendations

Area	Recommendation	Evidence
Vendor dependence	Reduce reliance on a few suppliers: qualify alternates for top categories and negotiate volume terms with the biggest vendors in return for their scale.	Top 10 vendors = 65.69% of purchases; Diageo = 16.30%
Inventory	Cut reorder quantities for vendors with turnover below 1 and run clearance or promotions on their excess stock.	$2.71M net unsold capital; ten vendors below 1.0 turnover
Pricing and promotion	Test promotions, shelf placement or price changes on low-sales, high-margin brands to lift volume.	Brands below $560 sales and above 64.97% margin
Bulk purchasing	Use larger orders where demand is steady to capture lower unit cost, but avoid over-buying slow movers.	Large-order unit price about 72% below small orders
Margin management	Review items sold at a loss; consider selective price increases or bundles for high-volume vendors with thinner margins.	Minimum gross profit -$52,002.78; top-quartile margin 31.18%
Logistics	Investigate freight cost outliers by vendor and shipment size.	Freight ranges from $0.09 to $257,032.07

9. Limitations and Next Steps

Limitations

- **Single-period snapshot:** there is no time-series view, so seasonality and trends are not captured.
- **Filtering effect:** several analyses use only items with positive profit and sales, so averages describe profitable, selling products and overstate typical margin.
- **Margin definition:** profit margin is calculated on purchase cost, which gives higher values than the more common margin on sales.
- **Bulk-price comparison:** unit price differs by product type as well as order size, so the 72% gap is not a pure volume discount.
- **Net unsold capital:** negative balances for fast sellers offset positive balances, so the $2.71M total understates gross excess stock.
- **Correlation is not causation:** the relationships found describe patterns, not proven causes.

Next steps

- Add time-series analysis (monthly sales, seasonality and demand forecasting) to improve reorder decisions.
- Build a model to predict slow-moving items before they accumulate.
- Segment vendors and brands (for example by ABC analysis or clustering) for differentiated purchasing policies.
- Automate the pipeline so the dashboard refreshes on a schedule.

10. Tools and Technologies

Area	Tools
Data storage and querying	SQLite, SQL (joins, CTEs, aggregations)
Analysis	Python, Pandas, NumPy, SciPy (confidence intervals, Welch t-test)
Visualisation	Matplotlib, Seaborn, Power BI
Workflow	Jupyter Notebook, Python logging, Git and GitHub

11. Conclusion

The analysis turns separate transaction tables into one decision-ready view of vendor performance. It shows that purchasing is concentrated in a few vendors, that millions of dollars of working capital sit in slow-moving stock, and that the lowest-selling items earn the highest margins, which points to specific promotion, pricing and purchasing actions. Together, the notebooks and the Power BI dashboard give a repeatable way to monitor vendors and keep inventory in line with demand.